Problem Statement ID – FIN-05

Real-Time Climate Risk Scoring for MSMEs(Climora)

Team Name – 404 NOT FOUNDERS

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Detailed explanation of the proposed solution

Climora – Climate Intelligence. Smarter Decisions.
Climora is a climate-risk assessment desktop application that helps financial institutions evaluate how climate hazards affect MSMEs. It combines business profiles, weather data and financial vulnerability indicators to generate an explainable climate-risk score.

Climate Data Integration: Uses Open-Meteo to retrieve weather, rainfall and temperature indicators.
Risk Scoring Engine: Generates a 0–100 score using five dimensions: flood exposure, heavy rainfall, extreme heat, business vulnerability and financial vulnerability.

Interactive Dashboard: Displays risk categories, contributing factors and assessment details.

Mitigation Recommendations: Suggests practical measures to improve climate preparedness.

Assessment History: Uses SQLite to store business profiles and previous assessments.

Technology: Electron, React, Node.js, SQLite and Open-Meteo API.
Climora supports informed lending assessments and early intervention through transparent climate-risk insights. Its score is a prototype for decision support, not a validated credit rating or automatic loan decision.

How it addresses the problem

Climora uses real-time weather data to calculate climate-risk scores for MSMEs, identify major climate threats, and recommend mitigation measures. This helps banks make informed lending decisions and businesses improve climate preparedness.

Innovation and uniqueness of the solution

Climate-to-Financial Impact: Connects climate hazards with potential business disruption and financial losses.
Personalized Risk Scoring: Combines climate data, business details and financial indicators for MSME-specific assessments.
Actionable Risk Reduction: Provides explainable scores and practical recommendations to improve climate preparedness.

Beyond Weather Monitoring: Focuses on how climate hazards may affect business and financial resilience.
Business-Specific Vulnerability: Recognizes that MSMEs in the same location can face different risks due to their operations and finances.
Transparent Decision Support: Explains risk factors and recommends mitigation measures to support lenders and MSME owners.

Technologies to be used & Methodology

Programming Languages: TypeScript

Frontend: React.js

Desktop Framework: Electron

Backend: Node.js, Expressjs

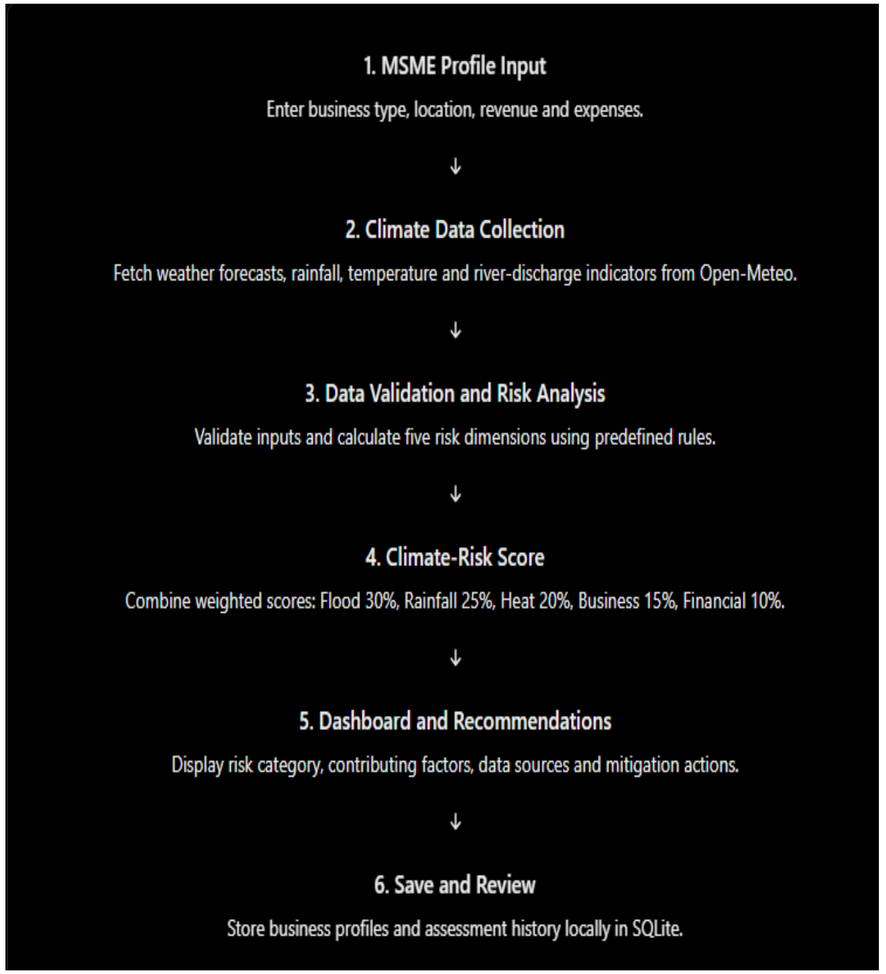
Database: SQLite (locally), PostgreSQL (Hosted)

Climate Data Integration: Open-Meteo Forecast, Flood and Geocoding APIs

Development Tools: Visual Studio Code, npm, Git and GitHub

Build Tools: Vite , Electron Builder

Authentication: Firebase



Feasibility and Viability

Technical Feasibility: Uses TypeScript, React, Electron, Node.js, SQLite and Open-Meteo APIs to build a desktop-based prototype.
Cost-Effectiveness: Uses open-source technologies and free weather-data access for the initial non-commercial prototype.
Practical Implementation: A rule-based scoring engine simplifies development and makes risk calculations transparent.
Scalability: The modular design can support additional climate indicators and datasets in future versions.
Real-World Viability: Helps banks assess climate-related MSME vulnerabilities and supports businesses in planning mitigation measures.

Impact and Benefits

Impact

Financial Institutions: Supports climate-aware lending and better-informed risk assessments.
MSMEs: Improves awareness of climate vulnerabilities and potential business disruptions.
Climate Resilience: Encourages early intervention and better business continuity planning.

Benefits

Risk Visibility: Identifies flood, heavy rainfall and extreme-heat risks.
Transparent Assessment: Explains risk scores and their contributing factors.
Practical Recommendations: Suggests measures to improve preparedness and financial resilience.
Assessment History: Stores previous assessments for review and comparison.
Responsible Decision Support: Complements traditional credit assessment without automatically approving or rejecting loans.

Research & References

Climate Risk Research

https://journals.plos.org/climate/article?id=10.1371/journal.pclm.0000644

Open-Meteo (Weather, Climate & Flood Data)

https://open-meteo.com/

Visual Crossing (Weather Data)

https://www.visualcrossing.com/weather-data/

Meteostat (Historical Climate Data)

https://dev.meteostat.net/

World Bank Climate Change Knowledge Portal

https://climateknowledgeportal.worldbank.org/download-data

World Bank Global Flood Exposure Dataset

https://datacatalog.worldbank.org/search/dataset/0062763/global-flood-exposure-gridded-exposure-headcounts-bycountry